AMRESHWAR SINGH

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EDUCATION

VIT Bhopal University, Bhopal, Madhya Pradesh – 466114 **Cumulative GPA: 7.94**
Bachelor of Technology in Computer Science, May 2027
With specialization in Artificial Intelligence & Machine Learning

TECHNICAL SKILLS

Languages & Backend: Python, REST APIs, FastAPI, SQLite, Google Cloud Platform
Distributed Systems & Databases: FAISS, ChromaDB, SQLite, multi-agent orchestration, parallel worker architectures
AI / ML Stack: LangChain, HuggingFace, Sentence-Transformers, Groq, Gemini 2.0, Vertex AI
Data & Analytics: NumPy, Pandas, Matplotlib, Seaborn, Tesseract, EasyOCR
Dev & Deployment: Git, Docker, Streamlit, Linux, Agent-based tooling

PROJECTS

Mental Wellness AI Agent | Multi-Agent Distributed Architecture | GitHub

- Built a low-latency crisis detection module with deterministic routing to emergency resources (988, Crisis Text Line), ensuring immediate escalation independent of conversational state.
- Architected a distributed 3-agent system (Analysis, Messaging, Orchestrator) with asynchronous message-passing, partitioned responsibilities, and well-defined inter-agent contracts.
- Designed a finite-state decision engine across 10 mental states × 4 intensity levels, dynamically generating personalized 7-day therapeutic programs from real-time user data.
- Used Vertex AI Memory Store as a persistent cross-session store; integrated Google Calendar and Spotify APIs across 4 automated daily workflows.

General-Purpose RAG Pipeline | Scalable Retrieval & Data Engineering | GitHub

- Engineered a parallel batch-processing system for 100+ documents with SQLite-backed crash recovery, guaranteeing exactly-once processing semantics under interruption.
- Designed a hybrid retrieval algorithm combining FAISS ANN search, ChromaDB vector storage, and a two-stage reranking layer for high-precision retrieval over large document corpora.
- Built a modular 7-stage pipeline (ingestion → chunking → embedding → indexing → retrieval → reranking → generation) architected for horizontal scalability and component-level replaceability.
- Resolved non-digital document ingestion using dual OCR engines (Tesseract + EasyOCR) with overlapping chunking to preserve semantic context at chunk boundaries.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- Applied Machine Learning in Python — University of Michigan (Coursera)
- Microsoft Certified: Azure Data Fundamentals — Microsoft
- Generative AI in Action— IBM SkillsBuild

EXTRA-CURRICULAR

Entrepreneurship Cell, VIT Bhopal Jan 2024 – Aug 2025
Content Lead — Operations, Finance & Media

- Managed content writing and documentation across Operations, Finance, and Media; established guided entrepreneurship communities within the university.
- Authored sponsorship brochures and memos that secured over ₹50k in sponsorships.
- Led and coordinated a team of 8 members across two consecutive tenures.